



LTV REPAIR PROCEDURES

Relevant to: EV Teams

NASA SUITS
2025-26 Challenge

This document **will** be updated ahead of test week in May 2026.

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Notes:

This document will likely undergo several more revisions as we do dry runs in preparation for test week, this implies the potential inclusion of additional errors and procedures. NASA SUITS aims to have a physical activity box with various panels and mock sensors for the EV to troubleshoot and repair. This physical box is representative of the Lunar Terrain Vehicle (LTV) External Control Panel and will be referred to as the “LTV Task Board”. Each item within the LTV Task Board will have a unique procedure for troubleshooting and repair, this document contains each of those procedures. These procedures are to be retrieved by your system from the TSS and during Test Week, your assigned Design Evaluator will be the one performing the procedures. **Do NOT store these procedures**, only reference them for understanding how the challenge will proceed. A complete procedure list will be available in the TSS and shall be retrieved by your Artificial Intelligence Assistant (AIA) during Test Week.

Change Log:

- 1. Flesh out initial tentative LTV Repair procedures, add cover page, general formatting, write *Notes* and *Procedures* section, override example errors with planned errors -RE 2/17/2026
- 2. Combine RTH and NAV Restart into one procedure, make comms error relevant to physical box, rewrite ERM to be a prerequisite to anything else, describe error formatting, error notes, and provide examples -RE 3/3/2026
- 3. Reiterate all current errors to follow format and layout of current task board, refine language and steps for errors that were previously vague -RE 4/9/2026

Error Format & Interface

Each procedure will have an error code associated with it. These error codes will be physically displayed on a seven-segment display on the LTV Task Board as well as existing in the TSS. These errors will consist of 4 digits corresponding to the following format:

Criticality Class	Subsystem Identifier	Error Identifier	Error Identifier
1-4	0-9	0-9	0-9

The subsystem identifier digit corresponds to the following list:

ID	Subsystem
0	Pre-Flight Integration and Testing
1	Secondary Computer Human Interface
2	Scientific Instrumentation
3	Lighting
4	Communications
5	Autonomous Guidance & Navigation

6	Vehicular Mobility & Control
7	Thermal Regulation
8	Diagnostic & System Monitoring
9	Power Distribution

Example Errors

Error 4965: A high criticality error within the Power Subsystem, identifier 65

Error 3224: A moderate-to-high criticality error within the Scientific Instrumentation Subsystem, identifier 24

Error 2777: A moderate criticality error within the Thermal Regulation Subsystem, identifier 77

Error 1401: A very low criticality error within the Lighting Subsystem, identifier 01

Interpreting & Triageing Errors

- Both the criticality class and the subsystem identifier follow a lowest-to-highest priority format, where 0/1 is the lowest priority, and conversely, 9 is the highest priority.
 - The error identifiers are **NOT** indicative of any priority; they serve solely to distinguish different errors.
- The criticality class corresponds to the subsystem, meaning two errors with equal criticality class should be prioritized based on which subsystem they correspond to.
 - Ex: Error 2700 should be fixed before 2600, and Error 3900 should be fixed before Error 2900
 - Teams should use their AIA to determine this ordering, but will also have to triage large disparities, such as choosing which to fix between a higher critically on a lower priority subsystem or a lower criticality on a higher priority subsystem (Ex: Error 4200 vs. Error 1800)
 - In some scenarios, it may be better to fix the higher criticality error on the lower priority subsystem
 - Teams are encouraged to use their AIA to triage errors in a non-linear fashion, considering exactly which subsystems are having the error and how those might affect the overall function of the LTV

Error Interface

These error codes will also exist in the TSS as a JSON object containing Booleans and the relevant procedures to fix these errors, please check the TSS documentation to reference the format of this JSON. This is to allow your system to assess which errors are present and simultaneously access the procedures to fix them.

Procedures

The primary task during the test session for EV Teams will be to repair the LTV systems that were damaged during the loss of communication. The following procedures will detail how to Exit Recovery Mode (ERM), Restart the navigation (NAV) system, as well as repair the LTV for each of the errors that could be indicated from the TSS. These errors will vary in criticality, and as per the [25-26 Mission Description](#), some fixes may be omitted if the time remaining is low. Subsequently, your AIA may be able to triage which errors to attempt to fix first. Additionally, as these procedures are iterated, they will become very verbose, and your AIA may shorten the steps in the procedures if able to do so while keeping critical information pertinent to the task.

Do NOT store these procedures, only reference them for understanding how the challenge will proceed.

Exit Recovery Mode (4800)

1. Investigate for prominent physical damage to any components before proceeding. This may include wires with damaged insulation, dents or large cracks visible on components, or missing wires
2. Verify the LTV is not stuck by confirming the wheels are not wedged in divots, rocks, and are otherwise intact.
3. Locate the power distribution and diagnostic (PDD) component by identifying an irregular-shaped white housing with four power bus connections
4. On the PDD display, verify the vehicle is in recovery mode and not some other error state, this is indicated by the error code 4800 and a red light by the recovery switch, labeled “RECO”
 - a. If a different error is displayed, that error must be fixed prior to exiting recovery mode
5. Via the Power Override Panel for Subsystems, labeled “POPS”, move switches for “MPS”, “GNC”, and “CDH” to the “OFF” position
6. Locate and press the Software Fault Acknowledge button, labeled “SFA”
7. Move the Power Optimization Mode switch, labeled “POM”, to the “OFF” position, verify change by confirming a blue indicator light
8. Open the protective cover for the External Battery Disconnect switch, labeled “EBD”
9. Press and hold the Computer Software Reboot button, labeled “CSR”
10. While holding the CSR button, move the EBD switch to the “ON” position
11. Release the CSR button
12. Wait 5-10 seconds for the reboot to take effect. Successful reboot will be indicated by the green light on the software management box turning on.
13. Move the EBD switch to the “OFF” position and close the cover
14. You should now see a red blinking status light appear on the software management box.
15. Via POPS, move the switches for MPS, GNC, and CDH to the “ON” position
16. Open the protective cover for the CTRL switch on the software management box.
17. On the software management box, hold down the black ALT button while flipping the CTRL switch on which will also turn the red status light off.
18. Release the ALT button

19. To test the software reboot, hold down both the test switch on the software management box and the CSR button until the green status light blinks. If the green status light is not blinking after 10 seconds, announce green light not blinking.
20. Hold down the black ALT button while switching off the control switch. The green status light should now be on to signify successful reboot.
21. Close the protective cover of the CTRL switch.
22. Open the protective cover for the recovery switch, labeled “RECO”
23. Move the RECO switch to the “OFF” position and close the cover
24. Verify the LTV has exited recovery mode by confirming the error code 4800 no longer displays
25. Announce ERM success

NAV Restart & Manual Return to Home (4509)

1. Locate the NAV Control Panel by identifying a large white rectangular component with a green light in the top-left corner
2. Flip the primary control switch under the green light to the far left, entering “HAND” mode. Verify mode switch by confirming the indicator light turns blue
3. Identify the “LIDAR RESET” button
4. Flush and restart the LIDAR system by depressing this button for 5 seconds. The LIDAR button indicator will light red upon depressing this button. The green indicator light on the software management box will turn on upon successful LIDAR system flush.
5. Identify the “NAV RESET” button
6. Flush and restart the NAV System by depressing this button for 5 seconds. The NAV button indicator will light red upon depressing this button. The red indicator light on the software management box will turn on upon successful NAV system flush.
7. Enact the reset by moving the primary control switch to the “OFF” position, wait for 10 seconds
8. Move the primary control switch to “HAND” mode. Verify mode switch by confirming the indicator light is blue
9. Identify the autonomous systems indicators toggle switch labeled “ASITS”
10. Move the ASITS switch to the “ON” position, verify mode switch by confirming the indicator light is yellow and the remaining indicators are on
11. Identify the autonomous navigation system lock labeled “ANAV BLOCK”
12. Move this switch to the “ON” position, verify mode switch by confirming the indicator light is blue
13. Identify the manual return to home switch labeled “ANAV RTH”
14. Move this switch to the “ON” position, verify mode switch by confirming the indicator light is blue
15. Identify the autonomy confidence adjustment dial labeled “ACA”
16. Rotate this dial counterclockwise to manually reduce the confidence scale of the autonomous navigation algorithm
17. Move the “ANAV BLOCK” switch to the “OFF” position
18. Move the “ASITS” switch to the “OFF” position
19. Identify the autonomous communication algorithm switch, labeled “COMM”
20. Turn the switch to the “SEC” position, and verify mode switch by confirming the indicator light is blue and the status buttons on the software management box have turned off.

21. Move the primary control switch to the “AUTO” position, verify mode switch by confirming indicator light is green
22. Announce NAV Restart & Manual Return to Home steps complete

Backup Fuse Error (1969)

1. Locate the fuse box by identifying a large gray rectangular housing
2. Lift the lid of the enclosure, exposing a black protective barrier, ensure the lid locks in the upright position
3. Remove the fuse disconnect by pulling the exposed black handle located towards the top of the box, while holding the enclosure steady
4. Remove the protective barrier by pulling outward from the top, while holding the enclosure steady
5. Use the provided fuse-puller tool to safely remove both fuses and properly discard them
6. Locate the replacement fuses
7. Using the fuse-puller tool, place a new fuse in the grips of the tool
8. Insert the fuse into the empty slot, moderate force may be required
9. Repeat steps 7 and 8 for the other fuse
10. Replace the black protective barrier
11. Reinsert the fuse disconnect in the proper orientation
12. Close the lid of the enclosure by lifting slightly outwards and pulling down
13. Announce Backup Fuse Error steps complete

Poor Comms RSSI (3452)

1. Locate the antenna communications control component by identifying a square gray box with white turn knobs and a strip of indicator lights
2. Locate the phase control knob, labeled “PHS”, rotate this until maximum possible RSSI improvement is achieved
3. Locate the modulation control knob, labeled “MOD”, rotate until maximum possible RSSI improvement is achieved
4. Locate the amplitude control knob, labeled “AMP”, rotate until maximum possible RSSI improvement is achieved
5. If the green indicator has not been lit, you may rotate these controls until maximum RSSI strength has been achieved, this is indicated by the green indicator being lit
6. Now, you should see a red blinking status light appear on the white rectangular box, the software management box
7. Flip open the cover of the CTRL switch
8. On the software management box, hold down the black ALT button while flipping the control switch on which will also turn the red status light off
9. To finish resetting the software for the RSSI component, press the black ALT button and the red RESET button at the same time. You should see the red and green status lights both turn on upon successful reset
10. To test the software reboot, hold down the test switch on the software management box until the green status light blinks. If the green status light is not blinking after 20 seconds, announce green light not blinking

11. Hold down the black ALT button while switching off the CTRL switch. Close the cover on the CTRL switch. The green status light should now be on to signify successful reboot
12. Announce Poor Comms RSSI steps complete

Comms Reboot Required (2441)

1. Locate the power distribution and diagnostic (PDD) component by identifying an irregular-shaped white housing with four power bus connections
2. Via the Power Override Panel for Subsystems, labeled “POPS”, move the Command and Data Handling switch, labeled “CDH” to the “OFF” position
3. Wait 5 seconds
4. Move the CDH switch to the “ON” position
5. Locate the antenna communications control component by identifying a square gray box with white turn knobs and a strip of indicator lights
6. If maximum RSSI strength is not indicated, proceed with the following steps
7. Locate the phase control knob, labeled “PHS”, rotate this until maximum possible RSSI improvement is achieved
8. Locate the modulation control knob, labeled “MOD”, rotate until maximum possible RSSI improvement is achieved
9. Locate the amplitude control knob, labeled “AMP”, rotate until maximum possible RSSI improvement is achieved
10. If the green indicator has not been lit, you may rotate these controls until maximum RSSI strength has been achieved, this is indicated by the green indicator being lit
11. Now, you should see a red blinking status light appear on the white rectangular box, the software management box
12. On the software management box, hold down the black ALT button while flipping the control switch on which will also turn the red status light off
13. To finish resetting the software for the RSSI component, press the black ALT button and the red RESET button at the same time. You should see the red and green status lights both turn on upon successful reset
14. To test the software reboot, hold down the test switch on the software management box until the green status light blinks. If the green status light is not blinking after 10 seconds, announce green light not blinking
15. Hold down the black ALT button while switching off the control switch. The green status light should now be on to signify successful reboot
16. Announce Comms Reboot steps complete

Subsystem Power Bus Error (4968)

1. Locate the power distribution and diagnostic (PDD) component by identifying an irregular-shaped white housing with four power bus connections
2. Locate the Power Override Panel for Subsystems, labeled “POPS”

3. Via POPS, move switches for “MPS”, “GNC”, “VSI”, and “CDH” to the “OFF” position
4. Check for any loose bus cables
5. If a cable is deemed loose, fully remove the cable from its socket
6. Firmly reinsert the cable into its socket
7. If no cable is deemed loose, fully remove and reinsert each cable into the respective sockets
8. Via POPS, move switches for “MPS”, “GNC”, “VSI”, and “CDH” to the “ON” position
9. Wait 5-10 seconds to verify the error clears before proceeding
10. Announce Subsystem Power Bus steps complete

Dust Sensor Error (2235)

1. Locate the dust sensor by identifying a cylindrical white housing smaller than the palm of a hand
2. Rotate the sensor body counterclockwise to unscrew it from its base and stow properly
3. Locate the replacement sensor and twist the protective cap counterclockwise to remove it, you may put the protective cap on the broken dust sensor
4. Insert the replacement sensor into the base and rotate clockwise until firmly secured
5. Announce Dust Sensor steps complete

Scientific Power Distribution Error (4280)

1. Locate the Scientific Management box by identifying a small gray box with a single protruding indicator light
2. Verify this indicator light is illuminated green, an unlit indicator implies an upstream power distribution issue
3. Locate the Scientific Power Distribution Module by identifying a small black housing with electrical terminals and six indicators
4. Take note of any indicators that are illuminated and which side they appear on
5. Locate the power distribution and diagnostic (PDD) component by identifying an irregular-shaped white housing with four power bus connections
6. Via the Power Override Panel for Subsystems, labeled “POPS”, move the Mission Payload Subsystem switch, labeled “MPS” to the “OFF” position
7. To prevent damage from unexpected electrical surges during operation, remove the power bus from its socket before proceeding
8. Corresponding to the side that had a lit indicator, locate the fuse cartridge for that side and remove it from the component
9. Prepare and insert a replacement fuse cartridge into the empty section
10. Via POPS, reinsert the power bus
11. Move the MPS switch to the “ON” position
12. Verify the scientific management indicator is illuminated green and no indicators are illuminated on the power distribution module
13. Announce Small Fuse Box steps complete